



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to KDCMPU LTD.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

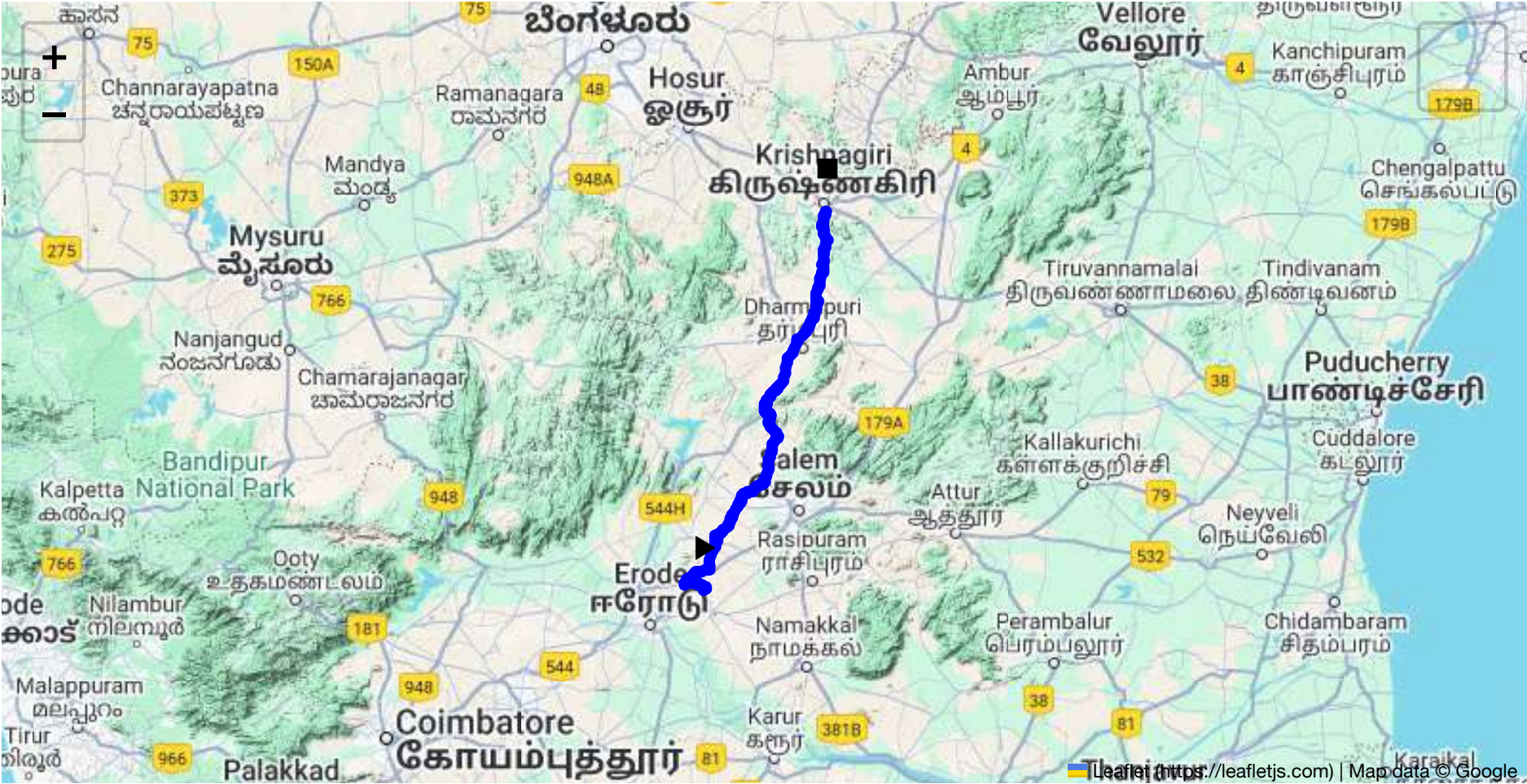
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 150.07 km
Estimated Duration: 2.9 hours
Adjusted Duration (Heavy Vehicle): 3.6 hours

Start: (11.4381, 77.8734)
End: (12.505128, 78.223618)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to G64F+3G8, Krishnagiri in Tamil Nadu navigates primarily through rural areas and small towns via Bhavani Main Road and State Highway 86, with a major waypoint in Salem, a significant urban center. The route spans approximately 150.07 kilometers.

Weather Conditions

Tamil Nadu generally experiences a tropical climate. Seasonal monsoons from June to September can lead to heavy rainfall, causing potential flooding and hazardous driving conditions. In months like April to June, high temperatures can contribute to heat-related vehicle issues and driver fatigue.

Traffic Patterns

The most congested segments are around Salem due to urban traffic and the junctions connecting to major highways. Peak congestion typically occurs from 8-10 AM and 5-8 PM. Outside urban areas, traffic is moderate, with occasional delays caused by local transport and agricultural vehicles.

Road Quality and Infrastructure

The road quality varies:

- **Bhavani Main Road to Padaiveedu:** Moderately maintained with occasional potholes.
- **SH 86 through Ayveli:** Generally good with proper signage.
- **Sections in Salem:** Urban roads are well-maintained but can be congested. Poorly lit rural stretches may present risks during night travel.

Alternative Routes

In case of emergencies, consider the following:

- Via NH 44: This is more direct from Salem to Krishnagiri but bypasses certain checkpoints. Can be used as an emergency route due to better infrastructure.
- Local routes such as NH 47 provide detours around affected segments, especially during weather-induced closures.

Local Regulations

Transporting hazardous materials requires permits and adherence to designated routes. Night travel may require additional clearances. It's crucial to check local transport regulations frequently as they can change.

Historical Incidents

The region has reported incidents of overturned lorries and accidents involving heavy vehicles primarily attributed to driver error in navigating sharp turns and speeding. No major hazardous material spills have been documented along this route recently.

Environmental Considerations

The route passes through several agricultural areas, so avoiding spillages is crucial to prevent groundwater contamination. Additionally, noise pollution should be minimized in populated areas.

Communication Coverage

Most of the route has good communication coverage, particularly near urban areas and major highways. Expect potential dead zones in rural segments, especially between Padaiveedu and Ayveli.

Emergency Response Times

- **Sangagiri to Salem:** Approx. 30-45 minutes
- **Salem region:** 15-30 minutes depending on traffic
- **Salem to Krishnagiri:** 45-60 minutes Delays should be expected in rural areas due to distance from major towns.

Overall Summary of Risk Assessment

This route is moderately risky for transporting hazardous materials due to variability in road quality and potential weather-related disruptions. Peak traffic in Salem and occasional communication dead zones pose additional challenges. Proper planning, route familiarity, adherence to regulations, and real-time traffic updates are crucial for minimizing risks. Emergency preparedness measures and alternative routing plans are advised to enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.55618, 77.89789	30 KM/Hr
11	Turn	Medium	11.55699, 77.89986	30 KM/Hr
12	Turn	High	11.58228, 77.90841	15 KM/Hr
13	Turn	High	11.58340, 77.90637	15 KM/Hr
14	Turn	Medium	11.59201, 77.90649	30 KM/Hr
15	Turn	Medium	11.63882, 77.94859	30 KM/Hr
16	Turn	High	11.64226, 77.95200	15 KM/Hr
17	Turn	Medium	11.64716, 77.95118	30 KM/Hr
18	Turn	Medium	11.67868, 77.96836	30 KM/Hr
19	Turn	Medium	11.70077, 77.97957	30 KM/Hr
20	Turn	High	11.70223, 77.97860	15 KM/Hr
21	Turn	Medium	11.71402, 77.99586	30 KM/Hr
22	Turn	Medium	11.71704, 77.99948	30 KM/Hr
23	Turn	Medium	11.71918, 78.01127	30 KM/Hr
24	Turn	Medium	11.72708, 78.02197	30 KM/Hr
25	Turn	High	11.73769, 78.03451	15 KM/Hr
26	Turn	High	11.73651, 78.03871	15 KM/Hr
27	Turn	Medium	11.74271, 78.04717	30 KM/Hr
28	Turn	Medium	11.86847, 78.08367	30 KM/Hr
29	Turn	Medium	11.87942, 78.08197	30 KM/Hr
30	Turn	Medium	11.92585, 78.06507	30 KM/Hr
31	Turn	Medium	11.93704, 78.05402	30 KM/Hr
32	Turn	High	11.95818, 78.04978	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
33	Turn	Medium	11.95988, 78.04948	30 KM/Hr
34	Turn	Medium	11.96304, 78.05136	30 KM/Hr
35	Turn	High	11.96399, 78.05135	15 KM/Hr
36	Turn	High	11.96580, 78.04923	15 KM/Hr
37	Turn	Medium	11.96675, 78.04939	30 KM/Hr
38	Turn	Medium	11.96791, 78.05070	30 KM/Hr
39	Turn	Medium	12.19881, 78.18157	30 KM/Hr
40	Turn	Medium	12.22459, 78.19136	30 KM/Hr
41	Turn	Medium	12.25060, 78.19977	30 KM/Hr
42	Turn	Medium	12.25329, 78.20039	30 KM/Hr
43	Turn	Medium	12.29110, 78.19909	30 KM/Hr
44	Turn	Medium	12.39904, 78.21764	30 KM/Hr
45	Turn	Medium	12.40156, 78.21845	30 KM/Hr
46	Turn	Medium	12.40589, 78.22275	30 KM/Hr
47	Turn	Medium	12.45018, 78.21828	30 KM/Hr
48	Blind Spot	Blind Spot	12.50718, 78.22253	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
31	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
22	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
27	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
28	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
29	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
30	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



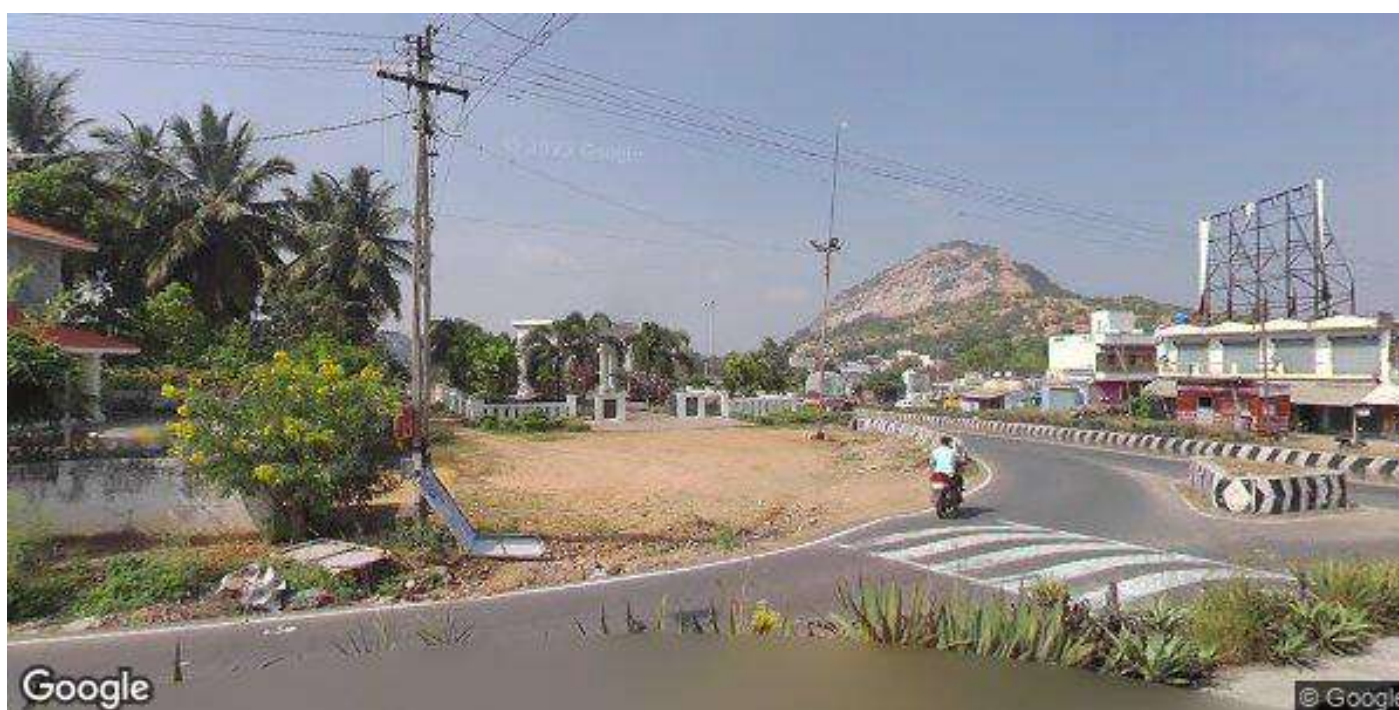
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55618, 77.89789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58340, 77.90637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59201, 77.90649



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63882, 77.94859



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64226, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64716, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70077, 77.97957



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71402, 77.99586



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71704, 77.99948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72708, 78.02197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45018, 78.21828



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.50718, 78.22253
